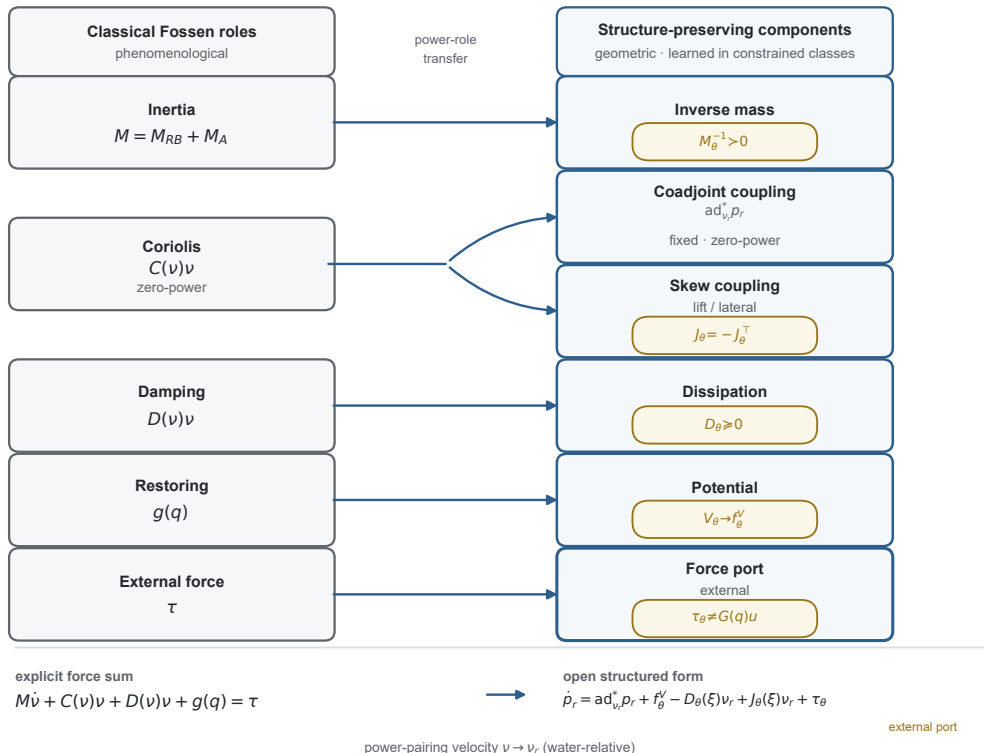




Mechanical-core power roles; current / actuator / depth are exogenous. The full vehicle–actuator–environment system is open — not a closed port-Hamiltonian system, and $\tau_\theta \neq G(q)u$.